Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 2 — Functions and graphs
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第2章 · 函数与图像
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

Figure 1 gives the complete graph of $y = f(x)$, formed from straight line segments joining $(-4, 1)$, $(-1, 4)$, $(2, -2)$ and $(5, 1)$. All four endpoints are included.

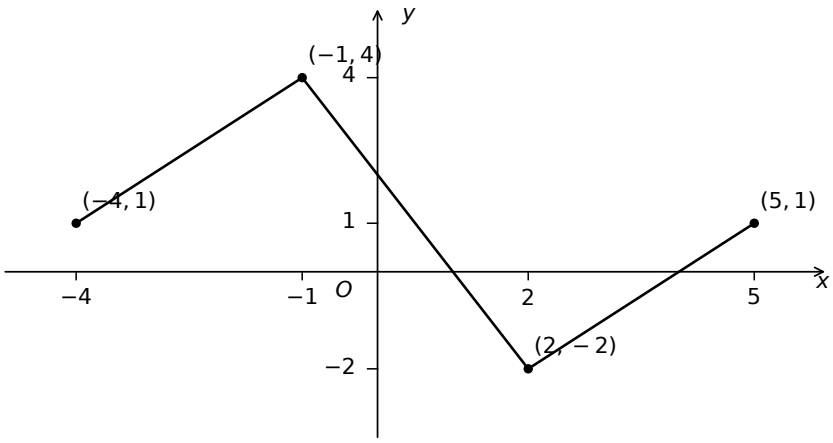


Figure 1

(a) Find $ff(2)$ and all solutions of $f(x) = 1$. (3)

(b) Let $g(x) = 2x - 1$ for real x. Find the domain and range of fg. (3)

(c) Sketch $y = |f(x)|$, marking every zero and every corner point. (3)

(Total for Question 1 is 9 marks)

The function $f(x) = 4 - (x - 1)^2$ has domain $[-1, 4]$. Define $g(x) = |f(x)|$ on the same domain, and $h(x) = f(|x|)$ wherever this is defined. Figure 2 shows the original graph.

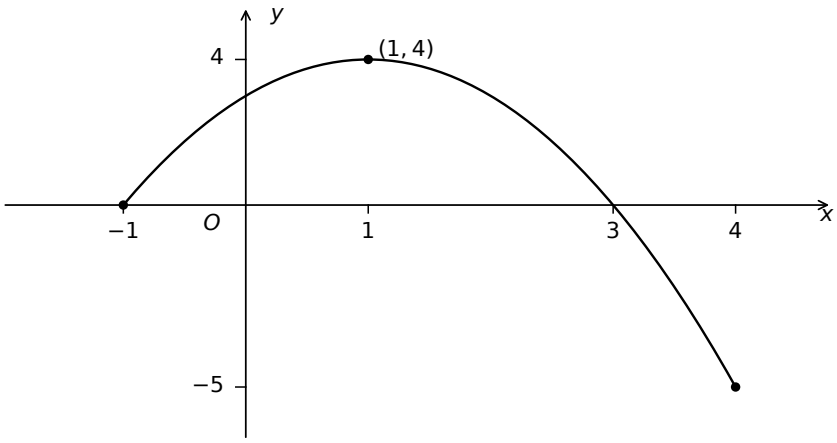


Figure 2

(a) Find the domain and range of h and sketch its graph, labelling its zeros and maximum points. (4)

(b) Find the range of g. (2)

(c) Find all k for which $y = g(x)$ and $y = k$ have exactly two distinct intersections. (3)

(Total for Question 2 is 9 marks)

An instrument maps a non-negative input x to the reading $f(x) = (ax + b)/(x + 2)$. Its calibration graph has horizontal asymptote $y = 3$ and passes through $(2, 2)$, as shown in Figure 3.

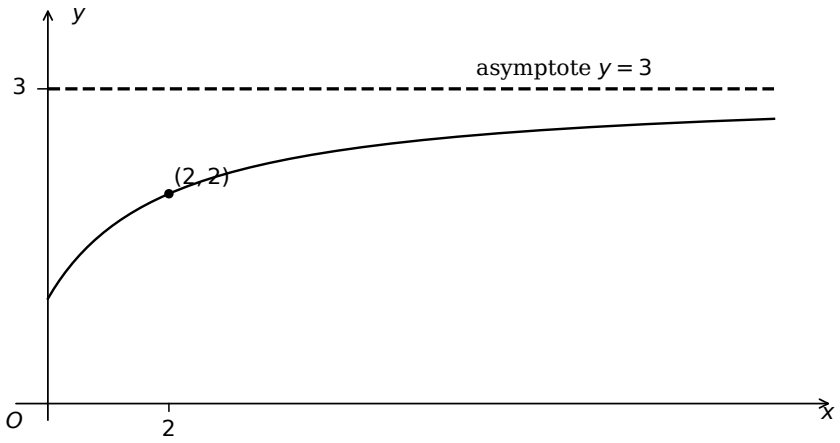


Figure 3

(a) Find a, b and write $f(x)$ in the form $3 - c/(x + 2)$. (3)

(b) Find the range of f and the inverse function, with its domain. (4)

(c) Find the input for which the reading is 0.2 below the limiting reading. Explain why the limiting reading itself is never attained. (3)

(Total for Question 3 is 10 marks)

4.

Let $f(x) = \sqrt{x - 1}$ for $x \geq 1$, and $g(x) = x^2 + 1$ for $x \geq 0$.

(a) Show that f and g are inverse functions, stating the domain of each composition. (4)

(b) The domain of g is now extended to all real numbers, while f is unchanged.

(i) Explain why g does not have an inverse function on this extended domain. (2)

(ii) Solve $fg(x) = 3$ for real x. (3)

(Total for Question 4 is 9 marks)

Figure 5 shows the complete graph of $f(x) = a|x - b| + c$ on $[-2, 6]$, where $a > 0$. Its vertex is $(2, -3)$ and its zeros are $-1, 5$. The two endpoints are included.

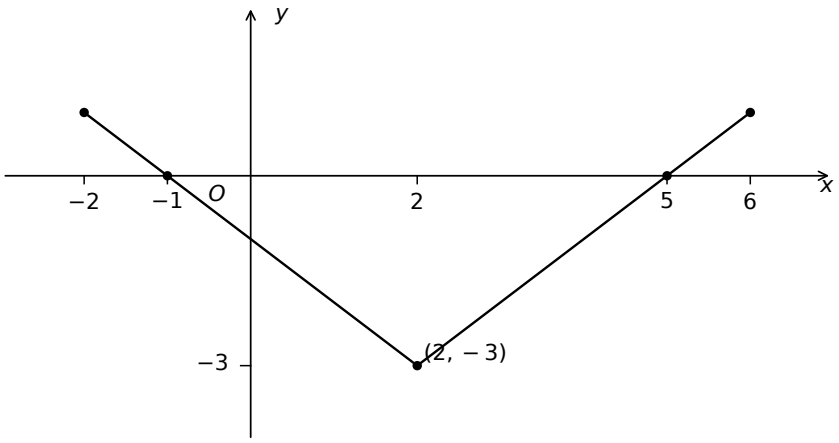


Figure 5

(a) Find a, b, c. (3)

(b) Find all k such that $f(x) + k = 0$ has two distinct solutions in the stated domain. (3)

(c) Solve $f(|x|) \geq 1$, respecting the domain of the composition. (3)

(Total for Question 5 is 9 marks)

A depth sensor in a tank of height 4 m gives reading $s = f(x) = (4 - x)^2$, where x m is the water depth and $0 \leq x \leq 4$. Figure 6 shows the calibration graph.

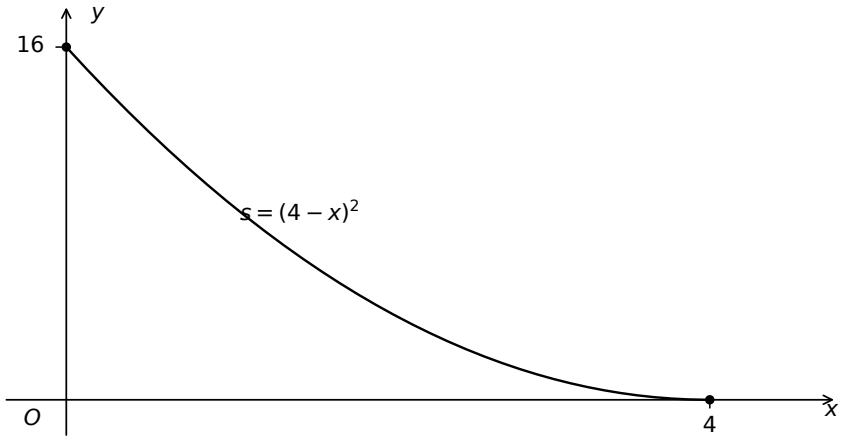


Figure 6

(a) State the range of the reading and find the depth corresponding to reading 9. (4)

(b) Find the inverse calibration function, with its domain. Explain the choice of square-root sign. (3)

(c) A second tank of height 8 m uses the sensor model $h(x) = f(x/2)$ for $0 \leq x \leq 8$.

(i) Find $h^{-1}(s)$, stating its domain. (3)

(ii) Find the depth corresponding to reading $s = 9$. (2)

(Total for Question 6 is 12 marks)

Let $f(x) = 1/(x-2)$ for $x > 2$, and $g(x) = \sqrt{x+1}$ for $x \geq -1$.

(a) Find $fg(x)$ and its domain. (3)

(b) Find $gf(x)$ and its domain and range. (4)

(c) Solve $fg(x) = 1$ and $gf(x) = \sqrt{3}$, giving each answer separately. (4)

(Total for Question 7 is 11 marks)

Figure 8 shows the complete graph of f on $[-2, 6]$. Its maximum is $f(0) = 5$, its minimum is $f(3) = -1$, and all points between these output values occur. Define $h(x) = -\frac{1}{2}f(2x) + 3$.

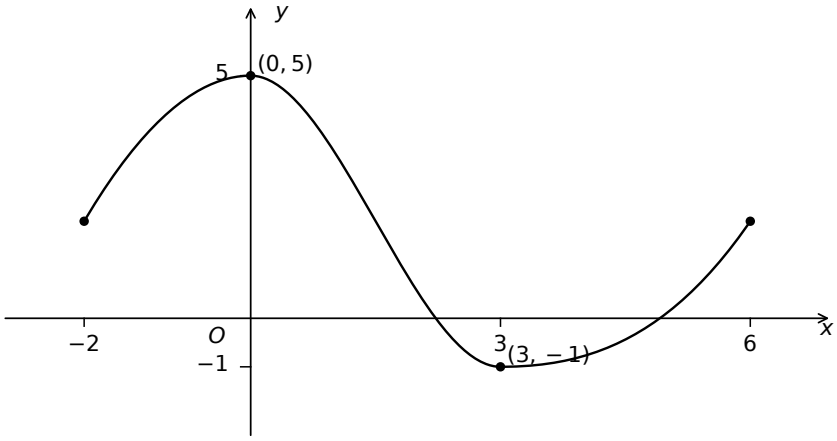


Figure 8

(a) Describe a sequence of transformations from the graph of f to that of h. (3)

(b) Find the domain, range and both extreme points of h. (3)

(c) Find the domain and range of $f(|x|)$. Explain why the range is not lost by discarding the negative-input part of f. (3)

(Total for Question 8 is 9 marks)

9. P3-C02-R09 · P3 · Chapter 2

Let $f(x) = (x+1)/(x-1)$ for real $x \neq 1$.

(a) Find f^{-1} with its domain, and hence find $ff(x)$. (4)

(b) Using this function, give a counterexample to the statement that a function graph and its inverse graph can intersect only on $y = x$. (3)

(c) Find the points where the common graph does meet $y = x$. (2)

(Total for Question 9 is 9 marks)

The function is defined by $f(x) = x + 2$ for $-2 \leq x \leq 1$, and $f(x) = 5 - x$ for $1 < x \leq 4$. Figure 10 shows the open and closed endpoints.

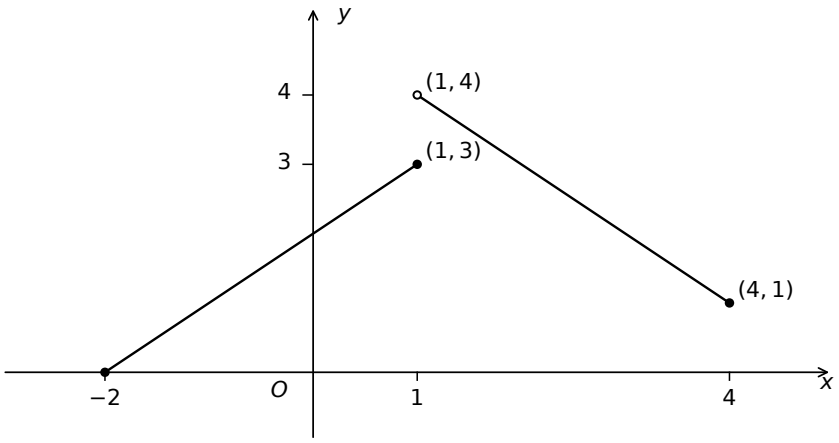


Figure 10

(a) Find the range of f and $ff(1)$. (3)

(b) Does f attain a greatest value? Explain using the endpoint at $x = 1$. (2)

(c) Restrict the domain to $[-2, 3/2]$. Find the inverse function in piecewise form, with its complete domain. (4)

(Total for Question 10 is 9 marks)

Let $f(x) = |x - 3|$ and $g(x) = |x| - 3$, both on the real numbers.

(a) Find $fg(x)$ and $gf(x)$, simplifying the outer absolute value where possible. (3)

(b) Solve $fg(x) = gf(x)$ for real x. (4)

(c) Use the input $x = 0$ to verify that the two composite functions are not identical. (2)

(Total for Question 11 is 9 marks)

Let $g(x) = 3 - \sqrt{x + 2}$ for $-2 \leq x \leq 7$. Figure 12 shows its graph.

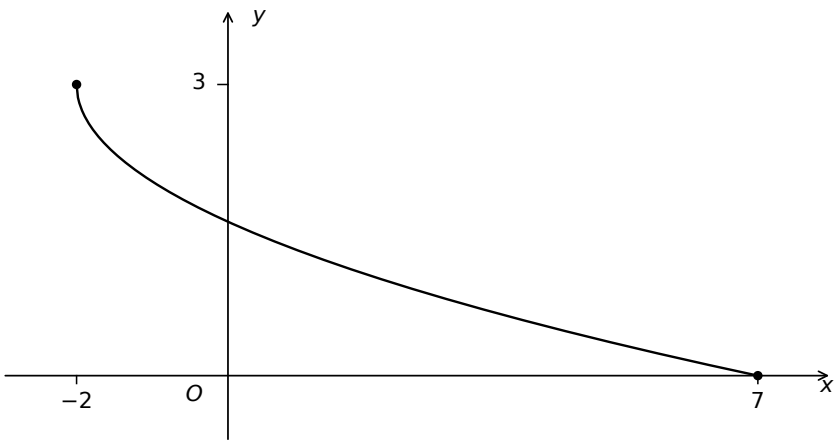


Figure 12

(a) Find its range and inverse function, with the domain of the inverse. (4)

(b) Solve $gg(x) = 1$ without expanding a nested square root. (3)

(c) Explain why gg is defined for every x in the original domain. (2)

(Total for Question 12 is 9 marks)

Let $f(x) = \sqrt{9 - x^2}$ on $[-3, 3]$, and $g(x) = x^2$ for real x.

(a) Find $fg(x)$ and its full domain and range. (4)

(b) Find $gf(x)$ and its full domain and range. (3)

(c) Solve $fg(x) = \sqrt{5}$ and $gf(x) = 5$, giving both solution sets. (2)

(Total for Question 13 is 9 marks)

The graph in Figure 14 is $y = f(x) = |x^2 - 4|$ on $[-3, 3]$. Define $h_k(x) = f(x) + k$ on this domain.

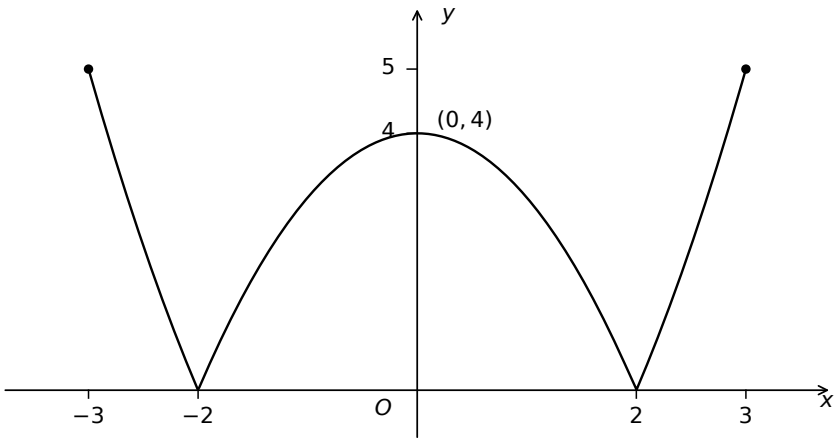


Figure 14

(a) Find the range of f and identify every point where the minimum occurs. (3)

(b) Find all k for which h_k has exactly four distinct zeros. (3)

(c) Find the value of k for which h_k has exactly three distinct zeros, and give these zeros exactly. (3)

(Total for Question 14 is 9 marks)

Let $q(x) = (x^2 - 9)/(x - 3)$ for real $x \neq 3$.

(a) Sketch the graph of q, showing the missing point. State its range. (3)

(b) Find $qq(x)$ with its complete domain and range. (4)

(c) Find q^{-1} with its domain. (2)

(Total for Question 15 is 9 marks)

Figure 16 shows $f(x) = x^3 - 3x$ on $[-2, 2]$. Its turning points are $(-1, 2)$ and $(1, -2)$, and its endpoints are $(-2, -2), (2, 2)$.

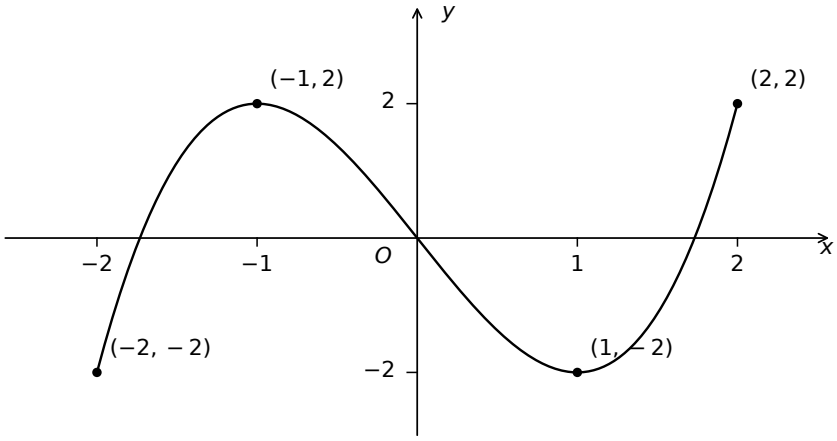


Figure 16

(a) Find all values of k for which $f(x) = k$ has exactly two distinct solutions. (3)

(b) Sketch $y = |f(x)|$, marking every zero and every point at height 2. (3)

(c) For which k does $|f(x)| = k$ have six distinct solutions? (3)

(Total for Question 16 is 9 marks)

A calibration is $f(x) = 18/(x+3)$ for $0 \le x \le 6$. A correction procedure adds 1 to a reading and then uses the inverse calibration. Its output is $H(x) = f^{-1}(f(x)+1)$ whenever this is defined.

(a) Find the range of f and f^{-1}, including its domain. (4)

(b) For the correction procedure $H(x) = f^{-1}(f(x)+1)$,

(i) Find the complete domain of H. (3)

(ii) Simplify $H(x)$ and find the input for which $H(x) = 1$. (4)

(Total for Question 17 is 11 marks)

Figure 18 gives a smooth graph $y = f(x)$ on $[-3, 4]$. Its only zeros are $-2, 1, 3$; its minimum is $(0, -2)$ and maximum is $(2, 4)$. The endpoint values are $f(-3) = 2$ and $f(4) = -1$. Define $h(x) = -2f(x/3) - 1$.

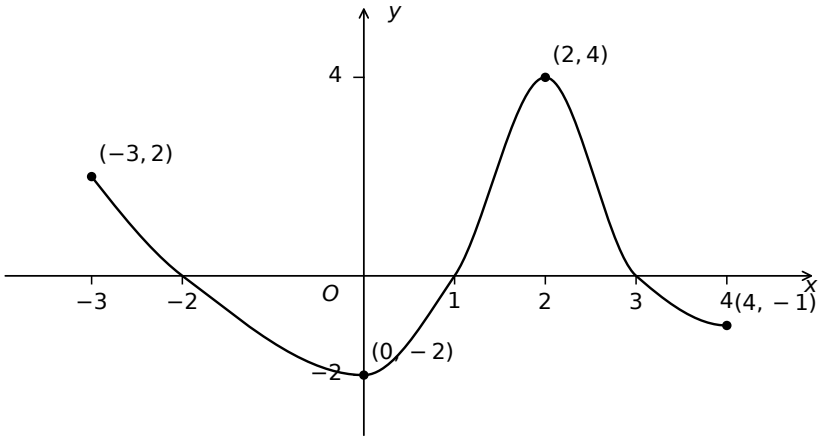


Figure 18

(a) Find the domain and range of h. (4)

(b) Find the maximum and minimum points of h. (3)

(c) Find all exact solutions of $h(x) = -1$ using the given graph information. (3)

(Total for Question 18 is 10 marks)

19. P3-C02-R19 · P3 · Chapter 2

Two depots lie at positions 1 and 4 km on a straight road. Travelling one kilometre to the first depot has cost 1 unit; the corresponding cost to the second depot is 2 units. A delivery point at x has total cost $F(x) = |x - 1| + 2|x - 4|$.

(a) Write F in piecewise form and sketch its graph. (3)

(b) Solve $F(x) \le 6$. (2)

(c) Restrict the delivery point to $x \ge 4$. Find the inverse cost function, with its domain. (3)

(Total for Question 19 is 8 marks)

Let $f(x) = \sqrt{16 - (x-1)^2}$ for $1 \leq x \leq 5$. Figure 20 shows its graph.

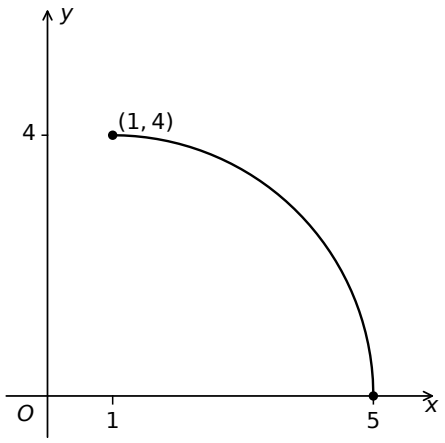


Figure 20

(a) Find f^{-1}, stating its domain and range. (4)

(b) Find all intersections of the two graphs exactly. Justify why none lies off $y = x$ for this particular pair. (4)

(Total for Question 20 is 8 marks)

The function $f(x) = |2^x - 4|$ is defined for all real x.

(a) Sketch the graph, labelling both intercepts and its horizontal asymptote, and state its range. (4)

(b) Find all values of k for which $f(x) = k$ has two distinct real solutions. (3)

(c) Solve $f(x) = 3$ exactly. (2)

(Total for Question 21 is 9 marks)

Let $f(x) = |1/x - 1|$ for real $x \neq 0$.

(a) Sketch the graph, showing all asymptotes and its zero, and state the range. (4)

(b) Find all k such that $f(x) = k$ has exactly two distinct solutions. (3)

(c) Solve $f(x) > 2$. (3)

(Total for Question 22 is 10 marks)

Let $f(x) = x^2 - 4x + 7$ for $x \geq 2$, and define $g(x) = f(x) - x$ on the same domain.

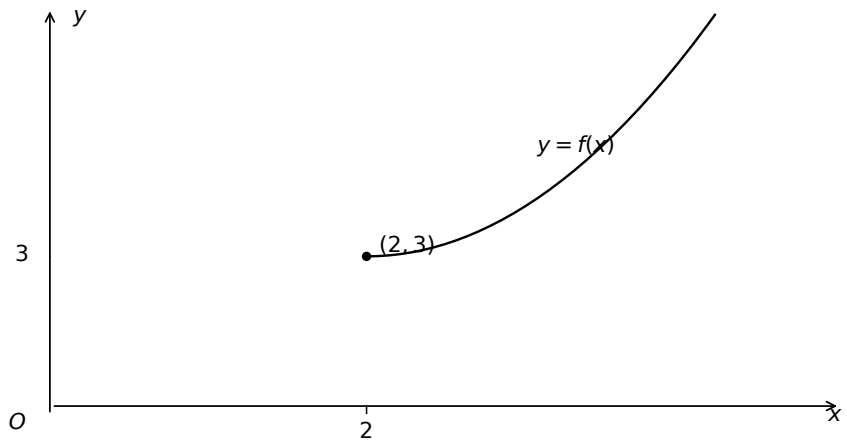


Figure 23

(a) Find f^{-1} with its domain. (3)

(b) Show that g is not one-to-one on its full domain, and find its range. (3)

(c) Find both inverses obtained by restricting g to its maximal one-to-one intervals within $[2, \infty)$. Include the domain of each inverse. (4)

(Total for Question 23 is 10 marks)

The complete graph of f is the two straight segments joining $(-2, -1)$ to $(0, 2)$ to $(2, -1)$, as shown in Figure 24.

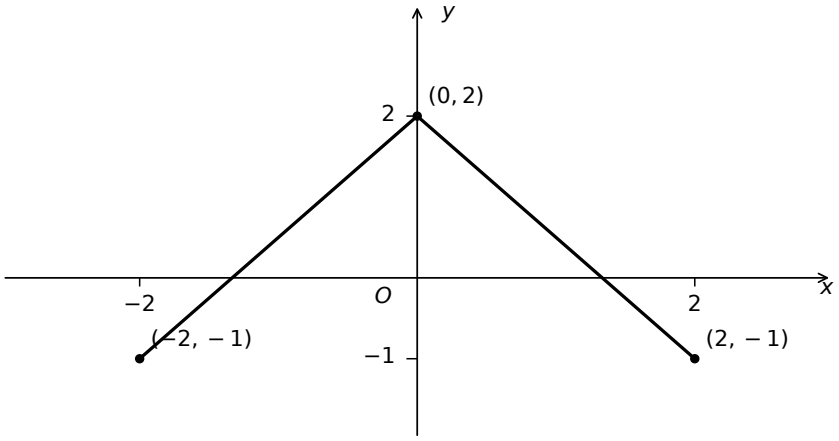


Figure 24

(a) Write an expression for f and state its range. (3)

(b) Restrict f to $[0, 2]$ and find the inverse of this restriction, including its domain. (3)

(c) Using the original full-domain f, find every solution of $ff(x) = 0$. (4)

(Total for Question 24 is 10 marks)

A two-step calibration applies $f(x) = (x+2)/(x+4)$ for $x \geq 0$, followed by $g(s) = 2s - 1$ for $1/2 \leq s < 1$. The complete calibration is $h = gf$.

(a) Find $h(x)$ and its range. (3)

(b) The complete calibration h is used to recover the original non-negative input from an output reading r.

(i) Find $h^{-1}(r)$ and state its domain. (3)

(ii) Find the input that gives reading $r = 1/4$. (2)

(c) A technician attempts to undo f first and then g. Explain the error using the reading $1/4$. (3)

(Total for Question 25 is 11 marks)

Let $f(x) = \dfrac{x-2}{x+1}$ for $x > 2$ and $g(x) = \dfrac{1}{x}$ for $x > 0$.

(a) Find the range of f and its inverse, including the domain of the inverse. (4)

(b) For the composite gf, find

(i) its expression and full domain, (3)

(ii) its range. (2)

(Total for Question 26 is 9 marks)

$f(x) = |x - 2| + 1$ is defined for $x \geq 2$. Define $h(x) = f(3 - x)$ wherever this is valid.

(a) Find f^{-1} and state its domain. (3)

(b) For h, find

(i) its domain and simplified expression, (3)

(ii) h^{-1} with its domain and range. (3)

(Total for Question 27 is 9 marks)

The function f is defined by $f(x) = x + 1$ for $0 \leq x < 2$, and $f(2) = 5$.

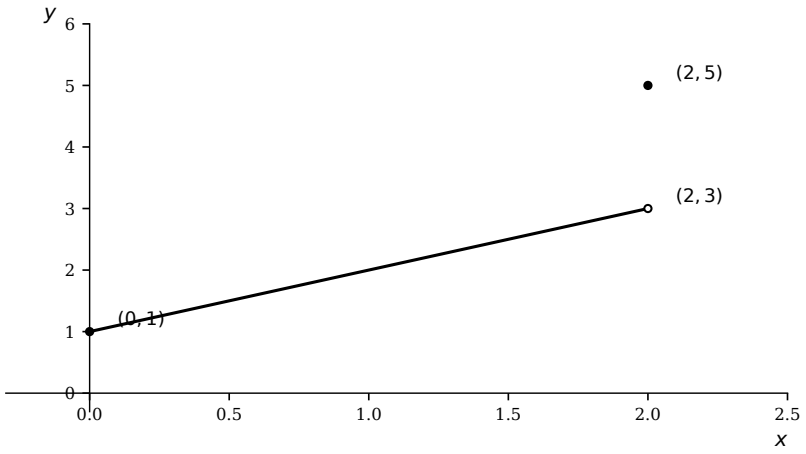


Figure 28

(a) State its range and determine whether it is one-to-one. (3)

(b) For the inverse function, find

(i) its expression on its complete domain, (3)

(ii) all x for which $ff(x)$ is defined. (3)

(Total for Question 28 is 9 marks)

$f(x) = \dfrac{2x+1}{x-2}$ for real $x \neq 2$.

(a) Find f^{-1} and show that $f^{-1}(x) = f(x)$ on their domains. (3)

(b) A translated function is $g(x) = f(x) + 3$.

(i) Find g^{-1} and state its domain, (4)

(ii) find all solutions of $g^{-1}(x) = 0$. (2)

(Total for Question 29 is 9 marks)

Let $f(x) = \sqrt{6 - x}$ for $x \leq 6$ and $g(x) = x^2$ for real x.

(a) Find $fg(x)$, with its full domain and range. (4)

(b) Now restrict fg to $0 \leq x \leq \sqrt{6}$.

(i) Find the inverse of this restriction, with its domain. (3)

(ii) find the exact intersection of this restricted graph and $y = x$. (2)

(Total for Question 30 is 9 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C02-R01 · 9 marks

(a) · 3 marks

$ff(2) = 3$; $x = -4, 1/2, 5$.

f(2)=−2，在第一线段f(−2)=3。水平线y=1与首点、中央线段x=1/2以及末点相交。

M1	按顺序从图读取复合值
A1	ff(2)
A1	三个有效横坐标

(b) · 3 marks

Domain $[-3/2, 3]$; range $[-2, 4]$.

要求2x−1∈[−4,5]，解得新区间。g在此区间恰好覆盖f整个定义域，所以复合值域与f相同。

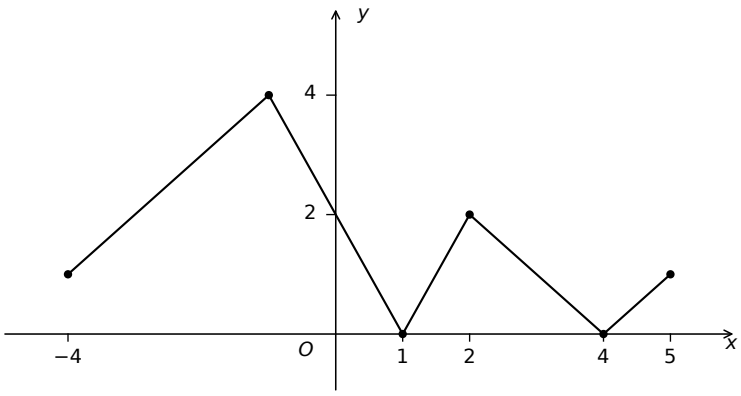
M1	内层输出必须落在外层定义域
A1	定义域
B1	完整值域及覆盖理由

(c) · 3 marks

Corner points $(-4, 1), (-1, 4), (1, 0), (2, 2), (4, 0), (5, 1)$; zeros $1, 4$.

保持横轴以上原图；把中央后段和最后前段的负值部分向上反射。中央线段零点1、末线段零点4，原低点(2,−2)变(2,2)。

M1	只反射原图负值部分
A1	两个零点
A1	其余全部角点和正确连线



Suggested sketch

Question 2

P3-C02-R02 · 9 marks

(a) · 4 marks

Domain $[-4,4]$, range $[-5,4]$; zeros $x=\pm 3$; maxima $(\pm 1,4)$; $h(0)=3$.

|x|只取原定义域中的[0,4]，再将这一半关于纵轴反射。不能把原图x<0的部分也保留。右端值−5仍是最小值。

M1	选取原图非负横坐标部分并反射
A1	定义域和值域
A1	两零点
A1	最大点及拼接点

(b) · 2 marks

$[0,5]$.

原函数最小−5、最大4且过轴；负值绝对值后能到5，故新最大值不是4。

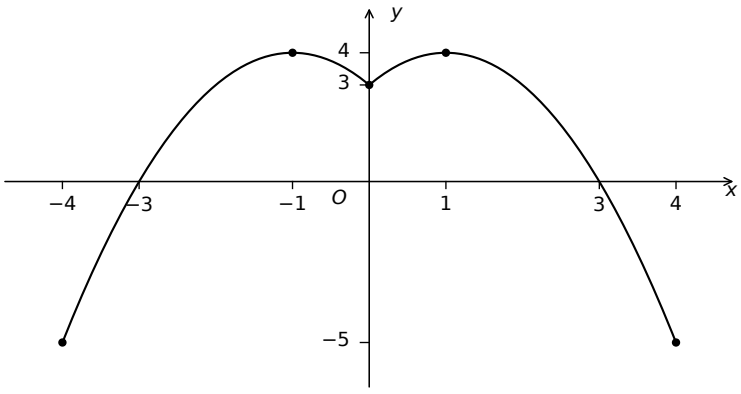
M1	比较原正最大值与负最小值绝对值
A1	完整值域

(c) · 3 marks

$k=0$ or $k=4$.

k=0时原零点−1、3两点；0

M1	按图分高度区间并处理极值
A1	k=0
A1	k=4且无额外值



Suggested sketch

Question 3

P3-C02-R03 · 10 marks

(a) · 3 marks

$a = 3, b = 2$; $f(x) = 3 - 4/(x + 2)$.

大x时分子分母首项比给a=3；点(2,2)给6+b=8。再用除法整理成3−4/(x+2)。

B1	从渐近线确定a
M1	代点求b
A1	参数与混合式

(b) · 4 marks

Range $[1, 3)$; $f^{-1}(x) = (2x - 2)/(3 - x)$, domain $1 \le x < 3$.

对x⩾0，4/(x+2)在(0,2]，故f在[1,3)。由y(x+2)=3x+2解出原输入，交换变量并用原值域作为逆函数定义域。

M1	用混合式求值域端点
A1	完整半开值域
M1	反解方程
A1	逆函数及其定义域

(c) · 3 marks

$x = 18$; $f(x) < 3$ for every finite $x \ge 0$.

与渐近线的差为4/(x+2)，令为0.2给x+2=20。对有限非负x这项严格为正，故无法正好输出3。

M1	用与极限的差建立方程
A1	输入18
B1	不能达到渐近线的代数理由

Question 4

P3-C02-R04 · 9 marks

(a) · 4 marks

$fg(x) = x$ on $[0, \infty)$; $gf(x) = x$ on $[1, \infty)$.

fg=√(x²)，因g的输入x⩾0可写x；gf=(√(x−1))²+1=x，需x⩾1。两个恒等式均必须带各自输入范围。

M1	正确计算fg并处理根号平方
A1	fg及范围
M1	正确计算gf
A1	gf及范围

(b) The domain of g is now extended to all real numbers, while f is unchanged.

(i) · 2 marks

$g(u) = g(-u)$ for every $u > 0$, so it is many-to-one.

正负同值输入输出相同，例如g(2)=g(−2)=5；逆映射将给同一输入两个输出，不能是函数。

M1	有效重复输出例子或一般关系
A1	非一一映射与无逆函数的解释

(ii) · 3 marks

$x = \pm 3$ because $fg(x) = |x|$.

√(x²)在实数域必须是|x|。扩展g允许负输入，因此−3不再受第一问域限制。

M1	识别全实数域下根号平方为绝对值
A1	两个解
B1	负解与域扩展的联系

Question 5

P3-C02-R05 · 9 marks

(a) · 3 marks

$a = 1, b = 2, c = -3.$

顶点直接给b=2、c=−3；代零点−1得3a−3=0。

B1	顶点确定b,c
M1	用一个零点定a
A1	参数

(b) · 3 marks

$-1 \le k < 3.$

解|x−2|=3−k。两不同根要求3−k>0；有限域距顶点左右最多都是4，故3−k⩽4。端点可取，顶点不能重复计为两解。

M1	用距离表达两根
M1	同时检查不同根和域端点
A1	正确半开区间

(c) · 3 marks

$x = -6$ or $x = 6.$

内层|x|必须落在[−2,6]，故|x|⩽6。再解||x|−2|⩾4：只有|x|⩾6可行，与前域结合得|x|=6。

M1	复合定义域
M1	解嵌套绝对值不等式
A1	仅两个端点

Question 6

P3-C02-R06 · 12 marks

(a) · 4 marks

$0 \le s \le 16$; depth 1 m.

4−x从4降至0，其平方取遍[0,16]；(4−x)²=9候选x=1、7，物理域只允许1。

M1	由物理域求输出范围
A1	完整值域
M1	解平方并检查物理域
A1	深度

(b) · 3 marks

$f^{-1}(s) = 4 - \sqrt{s}$, domain $[0, 16]$.

原模型中4−x≥0，因此4−x=√s。若取x=4+√s，除s=0外会超过 tank高度4；反函数域须是可实现读数范围。

M1	从原物理域选择根号正负
A1	正确逆函数
B1	逆函数定义域与解释

(c) A second tank of height 8 m uses the sensor model $h(x) = f(x/2)$ for $0 \le x \le 8$.

(i) · 3 marks

$h^{-1}(s) = 8 - 2\sqrt{s}$, $0 \le s \le 16$.

s=(4−x/2)²，由输入范围4−x/2≥0，反解x=8−2√s。

M1	正确反解及分支
A1	反函数
B1	定义域

(ii) · 2 marks

2 m.

代入逆校准8−2×3。

M1	代入同一传感器反函数
A1	深度

Question 7

P3-C02-R07 · 11 marks

(a) · 3 marks

$fg(x) = 1/(\sqrt{x+1} - 2)$; domain $x > 3$.

g输入先需x⩾−1，还要g(x)>2，故√(x+1)>2，最后得x>3。只排除分母为0还不够，因为f本身域为x>2。

M1	正确复合顺序
M1	检查外函数的完整输入域
A1	表达式和严格域

(b) · 4 marks

$gf(x) = \sqrt{1 + 1/(x-2)}$; domain $(2, \infty)$, range $(1, \infty)$.

f输入必须>2，其输出遍历正实数，全部可送入g；g对正输入的输出严格>1且无上界。

M1	正确复合
A1	域
M1	跟踪内层值域到外层
A1	值域

(c) · 4 marks

$x = 8$ and $x = 5/2$, respectively.

第一式给√(x+1)=3，x=8；第二式平方得1+1/(x−2)=3，x=5/2。分别检查其复合定义域。

M1	解第一复合并检查域
A1	第一解
M1	解第二复合并检查域
A1	第二解

Question 8

P3-C02-R08 · 9 marks

(a) · 3 marks

Horizontal scale factor $1/2$; reflection in the x-axis; vertical scale factor $1/2$; translation by $(0, 3)$.

内层2x把横坐标减半；外层−½把纵坐标反号并减半，最后纵向加3。

B1	横向尺度
B1	反射及纵向尺度
B1	最后的平移

(b) · 3 marks

Domain $[-1, 3]$; range $[1/2, 7/2]$; minimum $(0, 1/2)$, maximum $(3/2, 7/2)$.

新域由2x∈[−2,6]。负的纵向因子交换极大与极小，分别映射(0,5)和(3,−1)。

M1	正确输入范围缩放
A1	域和值域
A1	两个正确极值点

(c) · 3 marks

Domain $[-6, 6]$; range $[-1, 5]$. Both extreme outputs occur at non-negative inputs $0, 3$.

只用原f的[0,6]部分，然后反射到负横坐标；由于0和3已给最大、最小，且曲线连续，中间全部输出仍能达到。

M1	正确折叠定义域
A1	完整值域
B1	用两个极值位置解释

Question 9

P3-C02-R09 · 9 marks

(a) · 4 marks

$f^{-1}(x) = (x+1)/(x-1)$ on $x \neq 1$; $ff(x) = x$ for $x \neq 1$.

y(x−1)=x+1给x=(y+1)/(y−1)。原函数不能输出1，所以逆域排除1；它恰好是自身的逆。

M1	反解方程
A1	逆函数
B1	正确域
A1	合法复合恒等式

(b) · 3 marks

The two graphs coincide; for example $(2, 3)$ lies on both and is not on $y = x$.

第一问已得f=f逆，整幅图重合。以f(2)=3给出明确不在对角线的公共点，即构成反例。

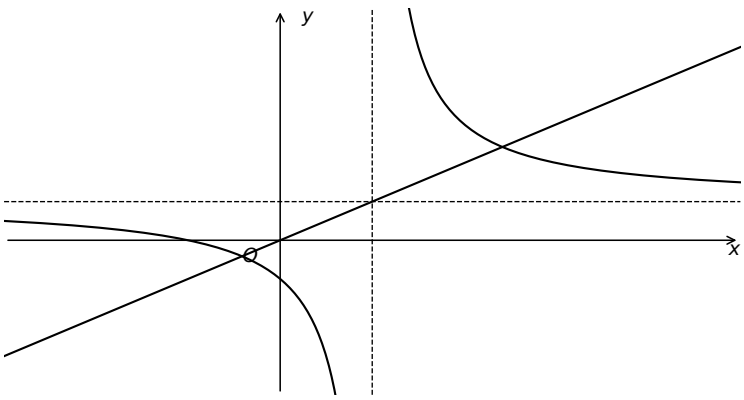
M1	由自身为逆说明两图重合
A1	有效离对角线公共点
B1	明确否定全称说法

(c) · 2 marks

$(1+\sqrt{2}, 1+\sqrt{2})$ and $(1-\sqrt{2}, 1-\sqrt{2})$.

解(x+1)/(x−1)=x得$x^2-2x-1=0$；两根均不等于1，坐标纵横相等。

M1	联立并解二次式
A1	两精确坐标且域有效



Suggested sketch

Question 10

P3-C02-R10 · 9 marks

(a) · 3 marks

Range $[0, 4)$; $ff(1) = 2$.

第一段取[0,3]；第二段取[1,4)，合并[0,4)。f(1)=3采用第一段，再算f(3)=2。

M1	正确处理两段端点
A1	完整半开值域
A1	复合值

(b) · 2 marks

No. Values approach 4 from below as $x \to 1^+$, but 4 is excluded and $f(1) = 3$.

靠近1右侧可输出比任何给定小于4的候选值更大的数，但4本身没有达到；不能把空心点当作函数值。

B1	未取得最大值
B1	空心端点及f(1)的解释

(c) · 4 marks

$f^{-1}(x) = x - 2$ for $0 \le x \le 3$; $f^{-1}(x) = 5 - x$ for $7/2 \le x < 4$.

第二段现在输入(1,1.5]、输出[3.5,4)，与第一段输出[0,3]不重叠，因此限制后是一一映射。两支分别反解并保留输出域中的缺口。

M1	检查两输出段不重叠
M1	分别反解
A1	两支公式
A1	两个区间及缺口

Question 11

P3-C02-R11 · 9 marks

(a) · 3 marks

$fg(x) = ||x| - 6|$; $gf(x) = |x - 3| - 3$.

f(g(x))=||x|−3−3|；g(f(x))=||x−3||−3，由于内层非负，可去掉外绝对值。

M1	正确两次代入顺序
A1	第一复合
A1	第二复合的化简

(b) · 4 marks

$x = -3$ or $x \geq 6$.

左边非负，故右边也须非负，先限x≤0或x≥6。x≥6时两式同为x−6，全部满足。x≤0再按−6分段：x≤−6无解，−6≤x≤0给x+6=−x，解−3。

M1	先限制右边非负的范围
M1	分别处理必要分段
A1	孤立解−3
A1	整段解x≥6

(c) · 2 marks

No: $fg(0) = 6$ while $gf(0) = 0$.

相等方程有无限多个解也不能说明两函数相同；只需在同一合法输入0处给不等输出。

B1	正确反例值
B1	非恒等的结论

Question 12

P3-C02-R12 · 9 marks

(a) · 4 marks

Range $[0,3]$; $g^{-1}(x)=(3-x)^2-2$ for $0\le x\le 3$.

根号从0增加到3，故输出从3减到0；反解√(x+2)=3−y，输出范围确保右侧非负，再平方。

M1	从原域求完整范围
A1	值域
M1	合法反解并平方
A1	逆函数及其域

(b) · 3 marks

$x=-1$.

先对外层用逆函数：g(x)=g逆(1)=2；再得x=g逆(2)=−1。核查g(−1)=2、g(2)=1。

M1	正确先撤销外层g
M1	再次用逆函数
A1	有效解

(c) · 2 marks

The range $[0,3]$ of the inner g is contained in the outer domain $[-2,7]$.

只有内层输出能全部送入外层，复合域才保持原域；此处确有[0,3]包含于[−2,7]。

B1	正确两个范围
B1	包含关系与全域可复合的解释

Question 13

P3-C02-R13 · 9 marks

(a) · 4 marks

$fg(x) = \sqrt{9 - x^4}$; domain $[-\sqrt{3}, \sqrt{3}]$; range $[0, 3]$.

g输出x^2要在f域$[-3,3]$，因非负，等价$x^2 \leqslant 3$；所得四次根式仍能由0取到3。

M1	正确复合
M1	外层域对x^2的限制
A1	定义域
A1	值域

(b) · 3 marks

$gf(x) = 9 - x^2$; domain $[-3, 3]$; range $[0, 9]$.

先做f需原域$[-3,3]$，其输出均可平方。平方抵消外根号，但不能据化简后的多项式把定义域扩展到实数。

M1	正确复合及化简
A1	保留原域
A1	完整值域

(c) · 2 marks

$x = \pm\sqrt{2}$ and $x = \pm 2$, respectively.

第一式平方得$x^4=4$，实数根$\pm\sqrt{2}$均在第一复合域；第二式得$x^2=4$，两根在第二域。

M1	分别解两个复合方程
A1	两解集及域核对

Question 14

P3-C02-R14 · 9 marks

(a) · 3 marks

Range $[0,5]$; minima $(-2,0),(2,0)$.

绝对值最小0在原二次式零点±2；端点x=±3给5，中央峰在x=0仅高4，不是全局最大。

M1	两种候选极大位置比较
A1	值域
A1	两个最小点

(b) · 3 marks

$-4<k<0$.

零点等价原图与水平y=−k相交；0<−k<4时四个交点，达到0或4时交点合并。

M1	用水平截线解释零点
M1	处理0与4两临界高度
A1	严格参数范围

(c) · 3 marks

$k=-4$; $x=0,\pm2\sqrt{2}$.

高度4触中央峰，再与两侧外支相交；解$|x^2-4|=4$给$x^2=0$或8，域内共有三个不同根。

M1	临界高度的交点合并
M1	解绝对值方程并去重
A1	k与三个精确根

Question 15

P3-C02-R15 · 9 marks

(a) · 3 marks

Line $y = x + 3$ with a hole at $(3, 6)$; range $\mathbb{R} \setminus \{6\}$.

约分得到x+3但原输入3仍不合法；在这条一一对应直线上删一个输入同时删对应输出6。

M1	正确约分但保留输入限制
A1	线与空心点
B1	值域缺6

(b) · 4 marks

$qq(x) = x + 6$, domain $\mathbb{R} \setminus \{0, 3\}$, range $\mathbb{R} \setminus \{6, 9\}$.

内层先排除x=3，还须q(x)≠3，即x+3≠3，另排除0。复合化简为x+6，所以失去输出6与9。

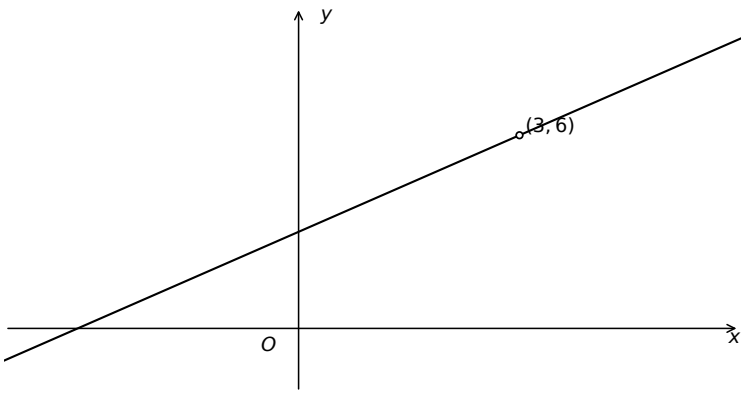
M1	同时检查两层定义域
A1	复合式
A1	完整域
A1	对应缺失值域

(c) · 2 marks

$q^{-1}(x) = x - 3,\ x \neq 6.$

反解y=x+3为原输入y−3，但逆输入不能用原函数缺失输出6。

M1	反解直线
A1	正确逆函数及其域



Suggested sketch

Question 16

P3-C02-R16 · 9 marks

(a) · 3 marks

$k = -2$ or $k = 2$.

水平高度在$(-2,2)$内给三交点；达到极大或极小时一对根合并为转折点，另有对侧端点，故两点。

M1	比较内部水平截线与临界高度
A1	两个k值
B1	不重复计相切根

(b) · 3 marks

Zeros $x = -\sqrt{3}, 0, \sqrt{3}$; height-2 points $(-2,2), (-1,2), (1,2), (2,2)$.

只把原图负值段关于横轴翻上去。零点由$x(x^2-3)=0$，端点和两转折点的绝对值都是2。

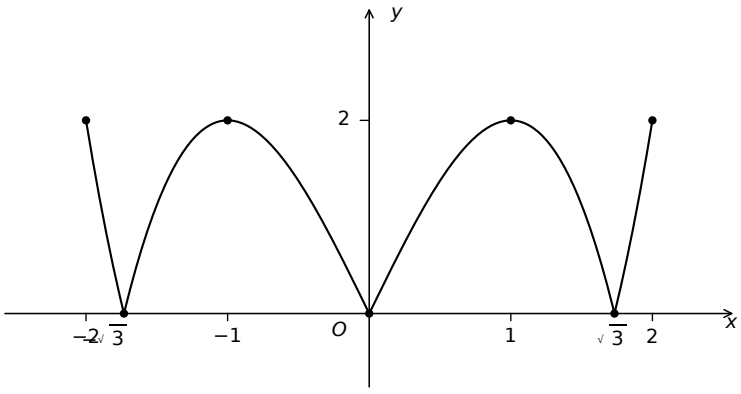
M1	正确反射负值部分
A1	三零点
A1	四个高度2的点与整体连线

(c) · 3 marks

$0 < k < 2$.

等价f=k或f=−k，每条非临界截线各三根；k>0使两组不同。k=0仅三根、k=2仅四根，故端点排除。

M1	拆成两条水平截线
M1	数清互不重合的根
A1	严格参数区间



Suggested sketch

Question 17

P3-C02-R17 · 11 marks

(a) · 4 marks

Range $[2,6]$; $f^{-1}(s)=18/s-3$ on $[2,6]$.

x+3在[3,9]内，倒数乘18得到输出[2,6]；反解18/(x+3)=s，得x=18/s−3。

M1	跟踪有限输入域到输出
A1	值域
M1	反解
A1	逆函数和域

(b) For the correction procedure $H(x)=f^{-1}(f(x)+1)$,

(i) · 3 marks

$3/5\le x\le 6$.

修正后读数必须满足2⩽f(x)+1⩽6。下限因f⩾2自动成立，上限要求18/(x+3)⩽5，故x⩾3/5，再与原域求交。

M1	检查修正读数落在逆函数定义域
M1	正确处理正分母不等式
A1	完整区间

(ii) · 4 marks

$H(x)=(15x-9)/(x+21)$; input $x=15/7$.

代逆函数得18/[18/(x+3)+1]−3，合并为(15x−9)/(x+21)。令其为1，得14x=30，且在有效域内。

M1	正确复合与通分
A1	最简式
M1	解输出方程并检查域
A1	有效输入

Question 18

P3-C02-R18 · 10 marks

(a) · 4 marks

Domain $[-9, 12]$; range $[-9, 3]$.

x/3必须在[−3,4]，所以横向放大3倍；纵输出负倍数−2交换范围两端，再减1，得[−9,3]。

M1	输入域变换
A1	定义域
M1	处理负倍数与平移
A1	值域

(b) · 3 marks

Maximum $(0, 3)$; minimum $(6, -9)$.

原最小(0,−2)映成(0,3)，原最大(2,4)映成(6,−9)；负号交换两者性质。

M1	变换两个点
A1	两坐标
B1	正确极值性质

(c) · 3 marks

$x = -6, 3, 9$.

h=−1等价f(x/3)=0，内层输入只能取原三个零点−2、1、3，故原输入各乘3。无需猜测图中其他高度的交点。

M1	目标高度转换到f的零水平
M1	正确反向横坐标映射
A1	三个解

Question 19

P3-C02-R19 · 8 marks

(a) · 3 marks

$F(x) = 9 - 3x$ for $x < 1$; $7 - x$ for $1 \le x \le 4$; $3x - 9$ for $x > 4$. Corners $(1, 6), (4, 3)$.

按1和4两地点分三段，逐个确定两个差值的符号。中段依然递减而不是水平，因为两运输权重不相同。

M1	在两个位置分段
A1	三个表达式和域
A1	两角点与正确三斜率

(b) · 2 marks

$1 \le x \le 5$.

左段要求x≥1故没有左开段内部解；中段全满足；右段3x−9≤6给x≤5，与x>4合并。

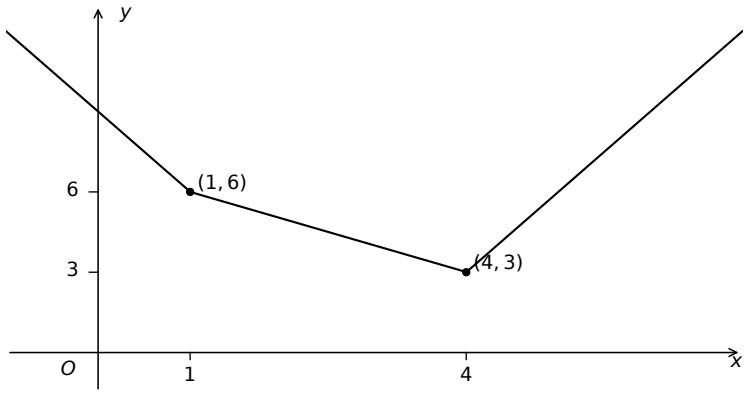
M1	各段解并结合对应范围
A1	两端与完整区间

(c) · 3 marks

$F^{-1}(c) = (c + 9)/3$ for $c \ge 3$.

该限制只留严格递增右支，成本最小3且无上界；从c=3x−9反解即可。

M1	选择一一右支并确定成本域
A1	逆函数
B1	正确逆域



Suggested sketch

Question 20

P3-C02-R20 · 8 marks

(a) · 4 marks

$f^{-1}(x) = 1 + \sqrt{16 - x^2}$ on $[0, 4]$, with range $[1, 5]$.

原横坐标≥1，所以x−1非负；反解时取正平方根。原输出从4降到0，因此逆输入域[0,4]。

M1	合法平方并解原输入
A1	选择正确分支
A1	逆函数
B1	逆域与范围

(b) · 4 marks

Only $((1 + \sqrt{31})/2, (1 + \sqrt{31})/2)$.

原图满足$(x-1)^2+y^2=16$，逆图满足$x^2+(y-1)^2=16$，相减得$y=x$。代回解$2x^2-2x-15=0$，负号根不在两个图的共同域；正号根合法。此理由来自本题两方程，不是普遍定理。

M1	写出两图方程并相减得y=x
M1	解二次方程
A1	正确精确坐标
B1	排除非法根与完整性

Question 21

P3-C02-R21 · 9 marks

(a) · 4 marks

Intercepts $(0,3),(2,0)$; asymptote $y=4$ as $x\to-\infty$; range $[0,\infty)$.

x<2时函数4−2^x，左端趋4但小于4；x≥2时2^x−4无限增长。左支的极限4不妨碍右支实际达到4。

M1	正确反射原指数负值支
A1	两截点
B1	渐近线方向
B1	完整值域

(b) · 3 marks

$0<k<4$.

拆成2^x=4+k与2^x=4−k。第一式k≥0时可解；第二式要求k<4；k=0两式同根，k=4第二式真数0无解。

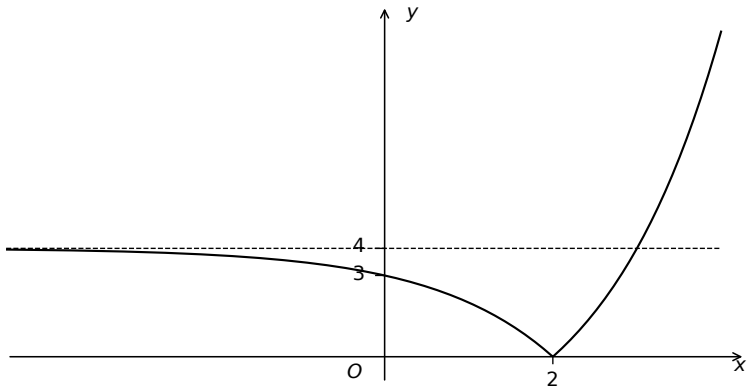
M1	两指数分支
M1	真数严格正的条件
A1	正确开区间

(c) · 2 marks

$x=0$ or $x=\log_2 7$.

分别由2^x=1或7，得到0与$\log_2 7$；两真数均正，不需P3指数微分。

M1	正确两绝对值分支
A1	两精确解



Suggested sketch

Question 22

P3-C02-R22 · 10 marks

(a) · 4 marks

Asymptotes $x = 0, y = 1$; zero $(1, 0)$; range $[0, \infty)$.

先画1/x−1，再把负值部分翻上；在x<0与x>1均为1−1/x，但分属不同分支；0

M1	正确三段形状
A1	两渐近线
B1	零点
B1	完整值域

(b) · 3 marks

$k > 0, k \neq 1$.

两候选输入为1/(1+k)与1/(1−k)。k=0时两候选重合；k=1时第二个不存在；其他正k均给两个合法非零输入。

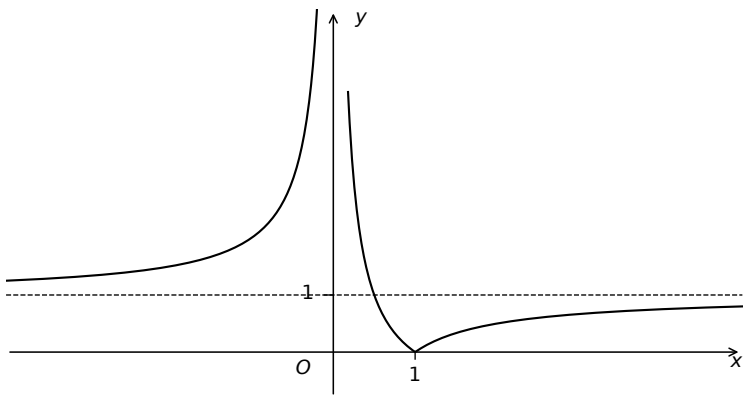
M1	解两个有理式分支
M1	检查重根和零分母例外
A1	完整参数集合

(c) · 3 marks

$-1 < x < 0$ or $0 < x < 1/3$.

按负横坐标、正横坐标小于1和大于1分段；负区1−1/x>2给−12给01时函数<1无解。

M1	完整分段并按分母符号解不等式
A1	负区
A1	正区与排除0



Suggested sketch

Question 23

P3-C02-R23 · 10 marks

(a) · 3 marks

$f^{-1}(x) = 2 + \sqrt{x-3}$ for $x \geq 3$.

f=(x−2)²+3，原横坐标在顶点右侧，输出⩾3，取正分支求逆。

M1	配方并限制根号分支
A1	逆函数
B1	逆域

(b) · 3 marks

$g(2) = g(3) = 1$; range $[3/4, \infty)$.

g=x²−5x+7=(x−5/2)²+3/4。定义域包含顶点两侧的2和3，所以一一性丢失；顶点仍在域内，给最小3/4。

M1	配方或有效重复值
A1	非一一的具体证据
A1	完整值域

(c) · 4 marks

On $[2, 5/2]$: inverse $(5 - \sqrt{4x-3})/2$ for $3/4 \leq x \leq 1$. On $[5/2, \infty)$: inverse $(5 + \sqrt{4x-3})/2$ for $x \geq 3/4$.

以顶点5/2分割原输入范围。两支反解为5/2±√(x−3/4)；左支仅有原输入[2,2.5]，所以其输出域是[.75,1]，不能把两逆函数都写成同一全半轴。

M1	找两段最大单调输入区间
M1	反解两个根号分支
A1	左支公式及有限域
A1	右支公式及半轴域

Question 24

P3-C02-R24 · 10 marks

(a) · 3 marks

$f(x) = 2 - 3|x|/2$ on $[-2, 2]$; range $[-1, 2]$.

顶点在(0,2)，左右斜率大小均3/2且向下，故表达式如此；图像连续覆盖−1到2。

M1	由图恢复斜率和绝对值结构
A1	正确函数
B1	完整值域

(b) · 3 marks

$f^{-1}(x) = (4 - 2x)/3$ on $[-1, 2]$.

右支是2−3x/2，一一递减；反解并使用这条支覆盖的输出区间[−1,2]。

M1	选择正确单调支
A1	逆式
B1	逆域

(c) · 4 marks

$x = \pm 4/9$.

外层f的零点是±4/3，但内层f只能输出[−1,2]，所以−4/3不可能。只解f(x)=4/3，得|x|=4/9。

M1	找外层的两候选零输入
B1	以内层值域排除−4/3
M1	解剩下的绝对值式
A1	两根

Question 25

P3-C02-R25 · 11 marks

(a) · 3 marks

$h(x) = x/(x+4)$; range $[0, 1)$.

f可写1−2/(x+4)，所以f输出[1/2,1)，与g输入域恰匹配。代入g得2(x+2)/(x+4)−1=x/(x+4)，输出从0到趋近1。

M1	检查两步的范围匹配
A1	复合简式
A1	完整值域

(b) The complete calibration h is used to recover the original non-negative input from an output reading r.

(i) · 3 marks

$h^{-1}(r) = \dfrac{4r}{1-r}$, $0 \le r < 1$.

h(x)=x/(x+4)，r(x+4)=x给x=4r/(1−r)；原输出范围作逆函数域。

M1	正确反解
A1	反函数
B1	定义域

(ii) · 2 marks

$4/3$.

代入(i)得(4×1/4)/(3/4)=4/3。

M1	代入完整逆校准
A1	输入

(c) · 3 marks

$h^{-1} = f^{-1}g^{-1}$, so g must be undone first. Reading $1/4$ is not even in the domain $[1/2, 1)$ of f^{-1}.

输出属于最后一步g，正确先反演g。若对1/4直接用f逆，输入已不在f可实现输出范围内，算式即便能形式代入也不是合法逆校准。

M1	正确反序规则
B1	用1/4与f逆域给具体非法性
A1	完整错误解释

Question 26

P3-C02-R26 · 9 marks

(a) · 4 marks

Range $(0,1)$; $f^{-1}(x)=\dfrac{x+2}{1-x}$ for $0<x<1$.

f=1−3/(x+1)，x>2时从0上升趋近1但不取端值；反解yx+y=x−2。

M1	范围端点与开闭
M1	反解
A1	逆表达式
B1	逆定义域

(b) For the composite gf, find

(i) · 3 marks

$gf(x)=\dfrac{x+1}{x-2}$, $x>2$.

f输入先需x>2，且其输出正，均可送入g；取倒数。

M1	正确复合顺序
A1	表达式
B1	定义域

(ii) · 2 marks

$(1,\infty)$.

gf=1+3/(x−2)，x>2时分式为任意正数，1不取到。

M1	正确改写与正值
A1	完整值域

Question 27

P3-C02-R27 · 9 marks

(a) · 3 marks

$f^{-1}(x) = x + 1,\ x \geq 1.$

原域使f=x−1且输出⩾1，反解原输入。

M1	用限制去绝对值
M1	反解
A1	逆函数及域

(b) For h, find

(i) · 3 marks

Domain $x \leq 1$; $h(x) = 2 - x$.

内部输入3−x⩾2给x⩽1；在此域内|1−x|=1−x，所以加1为2−x。

M1	内部输入约束
M1	合法去绝对值
A1	域与式

(ii) · 3 marks

$h^{-1}(x) = 2 - x$, domain $x \geq 1$, range $x \leq 1$.

h在x⩽1上输出⩾1，反解同一线性式，交换域和值域。

M1	反解
A1	逆函数
B1	域与值域

Question 28

P3-C02-R28 · 9 marks

(a) · 3 marks

Range $[1,3)\cup\{5\}$; one-to-one.

半开线段给[1,3)，孤立端点给5；各输出只对应一个输入，3、4不在值域。

M1	正确半开输出区间
A1	加孤立值
B1	一一性结论

(b) For the inverse function, find

(i) · 3 marks

$f^{-1}(x)=x-1$ for $1\le x<3$, and $f^{-1}(5)=2$.

每一段分别反解，保留孤立输入5，不填补3到5的空缺。

M1	连续分支反解
A1	孤立分支
B1	完整定义域

(ii) · 3 marks

$0\le x\le 1$.

需要x在原域且f(x)∈[0,2]；在半开线段f=x+1⩽2给x⩽1，x=2时输出5非法。

M1	复合输入条件
M1	排除特殊端点
A1	最终区间

Question 29

P3-C02-R29 · 9 marks

(a) · 3 marks

$f^{-1}(x) = \dfrac{2x+1}{x-2}, x \neq 2.$

yx−2y=2x+1给x(y−2)=2y+1，交换变量得同一式。

M1	正确反解
A1	表达式相同
B1	域

(b) A translated function is $g(x) = f(x) + 3$.

(i) · 4 marks

$g^{-1}(x) = \dfrac{2x-5}{x-5}, x \neq 5.$

撤销顺序先减3再应用f^{-1}：f(x−3)=[2(x−3)+1]/[(x−3)−2]。

M1	正确撤销顺序
M1	化简
A1	逆函数
B1	域

(ii) · 2 marks

$x = 5/2.$

分子2x−5=0，且x=5/2不等于逆域排除值5。

M1	分子零且分母不零
A1	解

Question 30

P3-C02-R30 · 9 marks

(a) · 4 marks

$fg(x) = \sqrt{6 - x^2}$; domain $[-\sqrt{6}, \sqrt{6}]$; range $[0, \sqrt{6}]$.

内部平方必须≤6，根式输出非负；在x=0取最高√6，两端为0。

M1	复合表达式
M1	内部约束
A1	定义域
B1	值域

(b) Now restrict fg to $0 \le x \le \sqrt{6}$.

(i) · 3 marks

Inverse $\sqrt{6 - x^2}$ on $[0, \sqrt{6}]$.

原输入非负使反解取正平方根；交换x和y得到同式，域由输出范围给出。

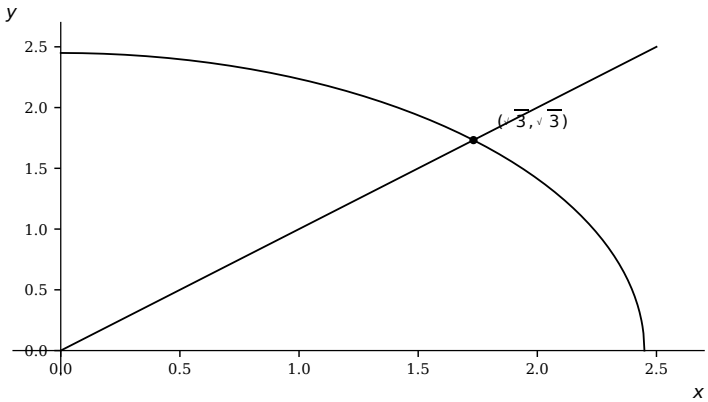
M1	正确平方反解
M1	分支选择
A1	逆函数及域

(ii) · 2 marks

$(\sqrt{3}, \sqrt{3})$.

平方$6-x^2=x^2$给$x^2=3$，限制$x\geqslant 0$只取$\sqrt{3}$。

M1	合法平方并求根
A1	唯一点



Suggested sketch